ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO TECHNICAL PROGRAMMING

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 3 Introduction to Technical Programming paper 1 · practice a.

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How to mark: each ✓ is one mark. Mark the logic, not small syntax errors. The ticks show where marks are given; accept other correct solutions.

SECTION A

QUESTION 1

1.1 C ✓
1.2 A ✓
1.3 D ✓
1.4 B ✓
1.5 D ✓
1.6 B ✓
1.7 C ✓
1.8 A ✓
1.9 B ✓
1.10 D ✓

(10 × 1) [10]

QUESTION 2

2.1 True ✓
2.2 False ✓
2.3 True ✓
2.4 False ✓
2.5 True ✓
2.6 True ✓
2.7 False ✓
2.8 True ✓
2.9 True ✓
2.10 True ✓

(10 × 1) [10]

QUESTION 3

3.1 C ✓
3.2 B ✓
3.3 F ✓
3.4 G ✓
3.5 D ✓
3.6 J ✓
3.7 A ✓

3.8 I ✓

3.9 E ✓

3.10 H ✓

(10 × 1) [10]

TOTAL SECTION A: [30]

SECTION B

QUESTION 4

4.1 setup() runs once when the board is powered or reset ✓ – used to set pin modes, start serial communication and initialise variables ✓; loop() runs repeatedly after setup() ✓ – contains the main program that reads inputs and controls outputs continuously ✓. (4)

4.2.1 INPUT_PULLUP keeps the pin HIGH when the button is not pressed ✓; pressing the button connects the pin to GND, so it reads LOW ✓. (2)

4.2.2 The LED on pin 9 is on while the button is held down ✓ and off when it is released ✓. (2)

4.2.3 Example:

```
bool ledOn = false;                          ✓ (state variable)
int lastState = HIGH;
void loop() {
  int state = digitalRead(button);
  if (state == LOW && lastState == HIGH) {   ✓✓ (detect a new press)
    ledOn = !ledOn;                          ✓ (toggle)
    digitalWrite(led, ledOn);                ✓
    delay(50);                               ✓ (debounce)
  }
  lastState = state;
}
```

(6)

4.3 Example:

```
void setup() {
  for (int p = 4; p <= 6; p++) pinMode(p, OUTPUT);   ✓✓
}
void loop() {
  for (int p = 4; p <= 6; p++) {                    ✓
    digitalWrite(p, HIGH);                          ✓
    delay(300);                                     ✓
    digitalWrite(p, LOW);                           ✓
  }
}
```

(6)

[20]

QUESTION 5

5.1 Example:

```
void setup() {
  pinMode(10, OUTPUT);                          ✓
}
void loop() {
  int val = analogRead(A0);                     ✓✓
  int bright = map(val, 0, 1023, 0, 255);       ✓✓
  analogWrite(10, bright);                      ✓
}
```

(6)

5.2.1 int reading = analogRead(A1); ✓ float volts = reading * 5.0 / 1024; ✓ float tempC = volts * 100; ✓✓ (since 10 mV = 1 °C) (4)

5.2.2 $V = 61 \times 5 \div 1\,024$ ✓ $= 0{,}30$ V $(0{,}298$ V) ✓; $T = 0{,}298 \div 0{,}01 = 29{,}79$ °C ≈ 30 °C ✓ (3)

5.3 PWM switches the output fully on and off very quickly ✓; the duty cycle (percentage of time on) controls the average power / brightness ✓; the voltage is always either 0 V or 5 V, never a steady in-between value (a filter would be needed for true analogue) ✓. (3)

[16]

QUESTION 6

6.1 Example:

```
if (dist > 30) {                    ✓
  forward();                        ✓
} else if (dist >= 15) {            ✓✓ (15 to 30 inclusive)
  slow();
} else {                            ✓
  turnAround();                     ✓
}
```

(6)

6.2 Example:

```
float average(int readings[]) {          ✓✓ (return type, array parameter)
  long sum = 0;                          ✓
  for (int i = 0; i < 5; i++) {          ✓
    sum += readings[i];                  ✓
  }
  return sum / 5.0;                      ✓ (float division)
}
```

(6)

6.3 delay() stops the whole program for the time given ✓, so the robot cannot read sensors or react during that time ✓; millis() lets the program check how much time has passed while it keeps running ✓, so several tasks can be done at the same time (non-blocking) ✓. (4)

[16]

QUESTION 7

7.1 Serial.begin(9600); ✓ (in setup) Serial.print("Distance: "); ✓ Serial.print(dist); ✓ Serial.println(" cm"); ✓ (4)

7.2 Any FOUR errors ✓ with corrections ✓: line 1 – missing semicolon → int led = 13; line 3 – pin set as INPUT → OUTPUT; line 6 – loops six times (0 to 5) → i < 5; line 7 – digitalwrite → digitalWrite (case); missing delay(500) after line 9, so the LED switches off too briefly to see the flash (8)

7.3 Any THREE: meaningful variable names ✓; comments explaining the code ✓; consistent indentation ✓; use constants instead of "magic numbers"; break code into functions (3)

7.4 Errors are easier to find in a small section ✓; you can check that each sensor or motor works before combining them ✓; it reduces the risk of damaging hardware or the robot behaving unsafely ✓. (3)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF MARKING GUIDELINE